

# Assembly & Machine Language Lab

## Assignment 2

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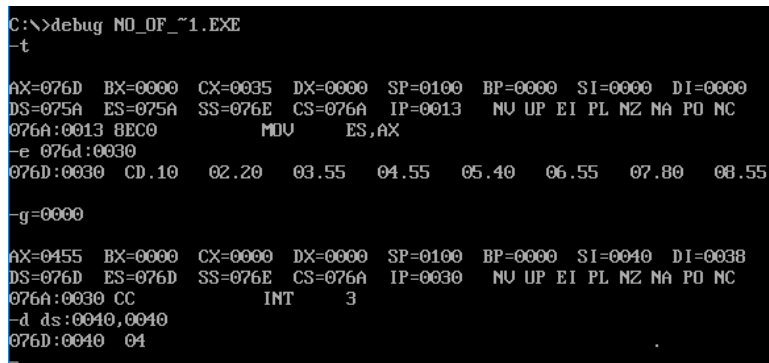
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1. Write an Assembly Language Program to count the number of occurrence of 55H in a string of eight data bytes. The starting address of string is DS: 0030H. Store the count value in DS:0040H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov ax,[si]
mov si,0040h
add ax,[si]
mov si,0050h
mov [si],ax
mov ah,0
adc ah,ah
add si,0002h
mov [si],ah
int 03h
mov ah,4ch
int 21h
main endp
end main
```

Output:



```
C:\>debug NO_OF_~1.EXE
-t
AX=076D  BX=0000  CX=0035  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076E  CS=076A  IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8EC0          MOV     ES,AX
-e 076d:0030
076D:0030 CD.10  02.20  03.55  04.55  05.40  06.55  07.80  08.55

-g=0000
AX=0455  BX=0000  CX=0000  DX=0000  SP=0100  BP=0000  SI=0040  DI=0038
DS=076D  ES=076D  SS=076E  CS=076A  IP=0030  NU UP EI PL NZ NA PO NC
076A:0030 CC          INT     3
-d ds:0040,0040
076D:0040 04
```

2. Write an Assembly Language Program to find out the location where 55H is placed in a string of eight data bytes. The starting address of string is DS: 0030H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov es,ax
mov di,0030h
mov ds,ax
mov si,0040h
mov cx,0008h
mov al,55h
cld
l2:  scasb
     jnz l1
     dec di
     mov [si],di
     add si,0002h
     inc di
     l1:  loop l2
int 03h
mov ah,4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug LOCATI~1.EXE
-t
AX=076D BX=0000 CX=0034 DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8EC0          MOV     ES,AX
-e 076d:0030
076D:0030 10.10  20.20  55.30  55.55  40.60  55.70  80.80  55.90
-g=0000
AX=0755 BX=0000 CX=0000 DX=0000 SP=0100 BP=0000 SI=0042 DI=0038
DS=076D ES=076D SS=076E CS=076A IP=002F  OU UP EI NG NZ NA PE CY
076A:002F CC          INT     3
-d ds:0040,0040
076D:0040 33

```

3. Write an Assembly Language Program to compare two strings. The first string is stored from memory location DS: 0030H and the second sting is stored from DS: 0040H. Consider that the first byte of both strings contain the number of bytes contained in that string. If both strings are found equal, then show a value FFFFH in address DS: 0050H, otherwise show 1111H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov es,ax
mov ds,ax
mov si,0030h
mov di,0040h

```

```

mov cl,[si]
mov ch,00h
cld
12:  cmpsb
    jnz 11
    loop 12
    mov ax,0ffffh
    jmp 13
11:  mov ax,1111h
13:  mov bx,0050h
    mov [bx],ax
    int 03h
mov ah,4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug COMP_2~1.EXE
-t
AX=076D BX=0000 CX=003A DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8EC0          MOV     ES,AX
-e 076d:0030
076D:0030 05.5    10.10   20.20   30.30   40.40

-e 076d:0040
076D:0040 05.5    10.10   20.20   30.30   40.40

-g=0000
AX=FFFF BX=0050 CX=0000 DX=0000 SP=0100 BP=0000 SI=0035 DI=0045
DS=076D ES=076D SS=076E CS=076A IP=0035  NU UP EI PL ZR NA PE NC
076A:0035 CC          INT     3
-d ds:0050,0051
076D:0050 FF FF

```

4. Write an Assembly Language Program to check if a string of five data bytes is palindrome or not. The string is stored from memory location DS: 0030H. If the string is found to be palindrome then place FFFFH in addresses DS: 0040H otherwise place 1111H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov di,0030h
mov ax,0000h
add al,[si]
dec al
add di,al
mov bl,02h
div bl
mov cl,al
mov ch,00h
inc si
cld

```

```

12:  mov al,[di]
    cmp al,[si]
    jnz 11
    inc si
    dec di
    loop 12
    mov ax,0ffffh
    jmp 13
11:  mov ax,1111h
13:  mov bx,0040h
    mov [bx],ax
    int 03h
mov ah,4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug PALIND~1.EXE
-t
AX=076E BX=0000 CX=004B DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076F CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8      MOV     DS,AX
-e 076e:0030
076E:0030 05.20  10.55  20.ff  30.55  40.20

-g=0000
AX=1111 BX=0040 CX=000F DX=0000 SP=0100 BP=0000 SI=0031 DI=004F
DS=076E ES=075A SS=076F CS=076A IP=0046  OV UP EI PL NZ NA PO NC
076A:0046 CC      INT     3
-d ds:0040,0041
076E:0040 11 11      ..
-

```

5. Write an Assembly Language Program to count the number of positive and negative numbers present in a series of eight data bytes. The starting address of the series is DS: 0040H. Store the count value of positive number in DS: 0040H and count value of negative number in DS: 0041H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0040h
mov cx,0008h
mov bx,0000h
13:  mov al,[si]
    rol al,01h
    jc 11
    inc bl
    jmp 12
11:  inc bh
12:  inc si
    loop 13
mov si,0040h
mov [si],bl
inc si

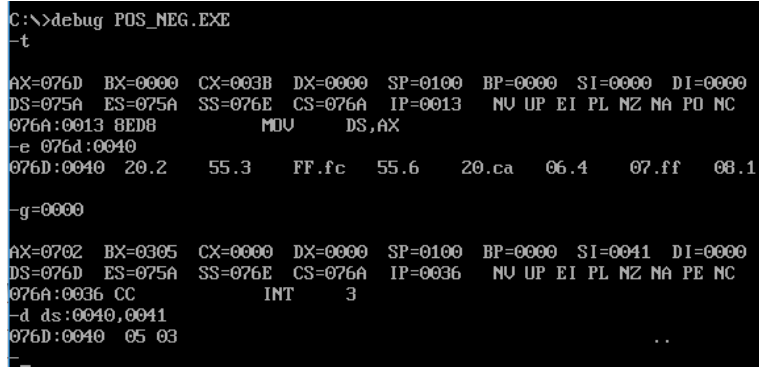
```

```

mov [si],bh
int 03h
mov ah,4ch
int 21h
main endp
end main

```

Output:



```

C:\>debug PDS_NEG.EXE
-t
AX=076D BX=0000 CX=003B DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013 NU UP EI PL NZ NA PO NC
076A:0013 8ED8 MOV DS,AX
-e 076d:0040
076D:0040 20.2 55.3 FF.fc 55.6 20.ca 06.4 07.ff 08.1
-g=0000
AX=0702 BX=0305 CX=0000 DX=0000 SP=0100 BP=0000 SI=0041 DI=0000
DS=076D ES=075A SS=076E CS=076A IP=0036 NU UP EI PL NZ NA PE NC
076A:0036 CC INT 3
-d ds:0040,0041
076D:0040 05 03

```

6. Write an Assembly Language Program to separate the odd and even numbers from a series of 7 data bytes. The starting address of the series is DS: 0030H. Store the even numbers from DS: 0040H and the odd numbers from DS: 0050H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov es,ax
mov bx,0030h
mov si,0040h
mov di,0050h
mov cx,0007h
l3:  mov al,[bx]
      ror al,01h
      jnc l1
      rol al,01h
      mov [di],al
      inc di
      jmp l2
l1:  rol al,01h
      mov [si],al
      inc si
l2:  inc bx
      loop l3
int 03h
mov ah,4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug EVEN_ODD.EXE
-t

AX=076D BX=0000 CX=003E DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8      MOV     DS,AX
-e 076d:0030
076D:0030 02.2    05.5    04.4    09.9    08.8    01.1    0A.a
-g=0000

AX=070A BX=0037 CX=0000 DX=0000 SP=0100 BP=0000 SI=0044 DI=0053
DS=076D ES=076D SS=076E CS=076A IP=0039  NU UP EI PL NZ NA PO NC
076A:0039 CC      INT     3
-d ds:0040,0043
076D:0040 02 04 08 0A      ....
-d ds:0050,0052
076D:0050 05 09 01      ...

```

7. Write an Assembly Language Program to convert an 8-bit number stored in DS: 0030H into its equivalent ASCII value. Store the converted code from DS: 0050H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov al,[si]
mov ah,al
mov cl,04h
and al,0fh
cmp al,0ah
jc l1
add al,07h
l1:  add al,30h
mov bx,0050h
mov [bx],al
mov al,ah
and al,0f0h
rol al,cl
cmp al,0ah
jc l2
add al,07h
l2:  add al,30h
      inc bx
      mov [bx],al
int 03h
mov ah,4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug ASCII_~1.EXE
-t
AX=076E BX=0000 CX=0043 DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076F CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8      MOV     DS,AX
-e 076e:0030
076E:0030 A2.b

-g=0000
AX=0B30 BX=0051 CX=0004 DX=0000 SP=0100 BP=0000 SI=0030 DI=0000
DS=076E ES=075A SS=076F CS=076A IP=003E  NU UP EI PL NZ NA PO NC
076A:003E CC      INT     3
-d ds:0050,0050
076E:0050 42      B

```

8. Write an Assembly Language Program to find out the square root of a number stored in DS: 0030H. Store the result in DS: 0040H.

```

dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov al,[si]
mov bl,01h
mov cl,00h
12:  sub al,bl
     js 11
     inc cl
     add bl,02h
     jmp 12
11:  add al,bl
     mov si,0040h
     mov [si],cl
int 03h
mov ah,4ch
int 21h
main endp
end main

```

Output:

```

C:\>debug SQRT.EXE
-t
AX=076D BX=0000 CX=0035 DX=0000 SP=0100 BP=0000 SI=0000 DI=0000
DS=075A ES=075A SS=076E CS=076A IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8      MOV     DS,AX
-e 076d:0030
076D:0030 10.10

-g=0000
AX=0700 BX=0009 CX=0004 DX=0000 SP=0100 BP=0000 SI=0040 DI=0000
DS=076D ES=075A SS=076E CS=076A IP=0030  NU UP EI PL ZR AC PE CY
076A:0030 CC      INT     3
-d ds:0040,0040
076D:0040 04

```

9. Fibonacci series is defined as:

$F(i) = F(i-1) + F(i-2)$ ; for all  $i > 2$  with  $F(1) = F(2) = 1$

Write an Assembly language Program to generate the first ten elements of this sequence and store them from DS: 0030H.

```
dosseg
.model small
.stack 100h
.data
.code
main proc
mov ax,@data
mov ds,ax
mov si,0030h
mov al,01h
mov bl,01h
mov cl,10h
mov dl,00h
l1:  mov [si],al
      mov dl,al
      mov al,bl
      add bl,dl
      inc si
      loop l1
int 03h
mov ah,4ch
int 21h
main endp
end main
```

Output:

```
C:\>debug FIB.EXE
-t
AX=076D  BX=0000  CX=0030  DX=0000  SP=0100  BP=0000  SI=0000  DI=0000
DS=075A  ES=075A  SS=076D  CS=076A  IP=0013  NU UP EI PL NZ NA PO NC
076A:0013 8ED8          MOV     DS,AX
-g=0000
AX=073D  BX=0018  CX=0000  DX=00DB  SP=0100  BP=0000  SI=0040  DI=0000
DS=076D  ES=075A  SS=076D  CS=076A  IP=002B  NU UP EI PL NZ AC PO CY
076A:002B CC          INT     3
-d ds:0030,0039
076D:0030 01 01 02 03 05 08 0D 15-22 37          ...."7
-
```